

IBM Watson OpenScale

GermanCreditRiskModelFRPROD Evaluation Report

October 04, 2025

Overview

Deployed model:

GermanCreditRiskModelFRPROD

Total test(s) failed

3

Report Details

Evaluated by: Armin Chanchian (armin.chanchian@s.utm.ro)
Report generated by: Armin Chanchian (armin.chanchian@s.utm.ro)
Report generated on: October 04, 2025 07:10:54 UTC

Model Details

Deployment ID: 2c40f0a6-6f2c-410b-83cb-71654f0c4413
Model name: GermanCreditRiskModelFRPROD
Model ID: be029f2b-bd33-42c0-a219-15848be410bc
Data type: Numeric/Categorical
Algorithm type: Binary classification
Number of explanations: 2

Training data details

Label column: Risk
Deployment prediction: predictedLabel
Training features: CheckingStatus, LoanDuration, CreditHistory, LoanPurpose, LoanAmount, ExistingSavings, EmploymentDuration, InstallmentPercent, Sex, OthersOnLoan, CurrentResidenceDuration, OwnsProperty, Age, InstallmentPlans, Housing, ExistingCreditsCount, Job, Dependents, Telephone, ForeignWorker

Metrics

Metric details

Summary

Deployed model
GermanCreditRiskModelFRPROD

Model ID
be029f2b-bd33-42c0-a219-15848be410bc

Metric
Quality

Alerts
5
RED BREACH

Summary

Number of metrics: 13
Alerts: 5

Statistics

True positive rate (TPR)

Status: **RED BREACH**
Score: 0.50
Threshold: 0.80

Area under ROC

Status: **RED BREACH**
Score: 0.73
Threshold: 0.95

Precision

Status: **GREEN**
Score: 0.85
Threshold: 0.80

Matthews correlation coefficient

Status: **GREEN**
Score: 0.54
Threshold: - -

F1-Measure

Status: **RED BREACH**
Score: 0.63
Threshold: 0.80

Accuracy

Status: **GREEN**

Metrics

Score: 0.80
Threshold: 0.80

Label skew

Status: GREEN
Score: 0.68
Threshold: - -

Gini coefficient

Status: GREEN
Score: 0.45
Threshold: - -

Logarithmic loss

Status: GREEN
Score: 0.47
Threshold: 0.80

False positive rate (FPR)

Status: GREEN
Score: 0.05
Threshold: 0.80

Area under PR

Status: RED BREACH
Score: 0.72
Threshold: 0.80

Recall

Status: RED BREACH
Score: 0.50
Threshold: 0.80

Brier score

Status: GREEN
Score: 0.49
Threshold: - -

Properties

Minimum sample size: 50
Evaluated records count: 200

Metric

Global explanation

Alerts

0

GREEN

Summary

Explanation method: Shap
Number of metrics: 1

Metrics

Alerts: 0

Statistics

Global explanation stability

Status: GREEN
Score: 0.996
Threshold: 0.800

Feature influence

LoanDuration: 0.037
Age: 0.036
CheckingStatus: 0.036
EmploymentDuration: 0.036
OwnsProperty: 0.030
LoanAmount: 0.019
Telephone: 0.014
ExistingCreditsCount: 0.010
CreditHistory: 0.008
OthersOnLoan: 0.008
SUM of 10 other features: 0.025

Properties

Evaluated records count: 50

Metric

Fairness

Alerts

4

RED BREACH

Summary

Number of metrics: 9
Monitored features: 2
Favorable outcome: No Risk
Unfavorable outcome: Risk
Alerts: 4

Age

Monitored groups: 18-25
Reference groups: 26-75
Alerts: 3

False positive rate difference

Status: RED BREACH
Score: 0.55
Threshold: 0.30

False negative rate difference

Status: GREEN
Score: -0.06

Metrics

Threshold: -0.30

Error rate difference

Status: GREEN

Score: 0.02

Threshold: -0.30

False omission rate difference

Status: GREEN

Score: -0.15

Threshold: -0.30

Disparate impact

Status: GREEN

Score: 102%

Threshold: 98%

Statistical parity difference

Status: GREEN

Score: 0.02

Threshold: -0.30

False discovery rate difference

Status: GREEN

Score: 0.00

Threshold: -0.30

Average odds difference

Status: RED BREACH

Score: 0.30

Threshold: 0.30

Average abs odds difference

Status: RED BREACH

Score: 0.30

Threshold: 0.30

Sex

Monitored groups: female

Reference groups: male

Alerts: 1

False positive rate difference

Status: GREEN

Score: 0.00

Threshold: -0.30

False negative rate difference

Metrics

Status: GREEN
Score: -0.08
Threshold: -0.30

Error rate difference

Status: GREEN
Score: -0.13
Threshold: -0.30

False omission rate difference

Status: GREEN
Score: -0.19
Threshold: -0.30

Disparate impact

Status: RED BREACH
Score: 92%
Threshold: 98%

Statistical parity difference

Status: GREEN
Score: -0.07
Threshold: -0.30

False discovery rate difference

Status: GREEN
Score: -0.14
Threshold: -0.30

Average odds difference

Status: GREEN
Score: 0.04
Threshold: -0.30

Average abs odds difference

Status: GREEN
Score: 0.04
Threshold: -0.30

Properties

Minimum sample size: 100
Evaluated records count: 100

Metric

Drift v2

Alerts

4

RED BREACH

Summary

Number of metrics: 4

Metrics

Alerts: 4

Model quality drift

Status: RED BREACH
Score: 0.19
Threshold: 0.05

Confidence drift

Status: RED BREACH
Score: 0.27
Threshold: 0.05

Feature drift

Status: RED BREACH
Score: 0.14
Threshold: 0.05

Highest feature impact scores

Age: 0.222
InstallmentPercent: 0.194
ExistingSavings: 0.174
OthersOnLoan: 0.139
LoanPurpose: 0.116

Prediction drift

Status: RED BREACH
Score: 0.06
Threshold: 0.05

Properties

Evaluated records count: 100

Metric

Model health

Alerts

0
GREEN

Summary

Number of metrics: 28
Alerts: 0

Statistics

Total records

Status: GREEN
Score: 100.00
Threshold: - -

Maximum payload size

Metrics

Status:	GREEN
Score:	0.16
Threshold:	--

Maximum record throughput

Status:	GREEN
Score:	695.01
Threshold:	--

Median api throughput

Status:	GREEN
Score:	593.71
Threshold:	--

Minimum api throughput

Status:	GREEN
Score:	392.71
Threshold:	--

Median record latency

Status:	GREEN
Score:	1.69
Threshold:	--

Average payload size

Status:	GREEN
Score:	0.16
Threshold:	--

Users

Status:	GREEN
Score:	1.00
Threshold:	--

Maximum api latency

Status:	GREEN
Score:	2.55
Threshold:	--

Maximum api throughput

Status:	GREEN
Score:	695.01
Threshold:	--

Total payload size

Status:	GREEN
Score:	15.62
Threshold:	--

Metrics

Total scoring requests

Status: GREEN
Score: 100.00
Threshold: - -

Median payload size

Status: GREEN
Score: 0.16
Threshold: - -

Median records

Status: GREEN
Score: 1.00
Threshold: - -

Median record throughput

Status: GREEN
Score: 591.32
Threshold: - -

Minimum api latency

Status: GREEN
Score: 1.44
Threshold: - -

Average record throughput

Status: GREEN
Score: 572.65
Threshold: - -

Maximum records

Status: GREEN
Score: 1.00
Threshold: - -

Minimum record latency

Status: GREEN
Score: 1.44
Threshold: - -

Average record latency

Status: GREEN
Score: 1.75
Threshold: - -

Median api latency

Status: GREEN

Metrics

Score: 1.68
Threshold: - -

Minimum records

Status: GREEN
Score: 1.00
Threshold: - -

Average api throughput

Status: GREEN
Score: 572.65
Threshold: - -

Maximum record latency

Status: GREEN
Score: 2.55
Threshold: - -

Average api latency

Status: GREEN
Score: 1.75
Threshold: - -

Minimum payload size

Status: GREEN
Score: 0.16
Threshold: - -

Minimum record throughput

Status: GREEN
Score: 392.71
Threshold: - -

Average records

Status: GREEN
Score: 1.00
Threshold: - -

Test summary

Tests passed

1

Number of evaluated records

0

Tests failed

3

Appendix

Quality Measures

Area under ROC
Area under PR
Accuracy
True positive rate (TPR)
False positive rate (FPR)
Recall
Precision
F1-measure
Logarithmic loss

Fairness measures

Fairness

Drift measures

Drop in accuracy
Drop in data consistency
Estimated accuracy
Base accuracy

Drift v2 measures

Output drift
Model quality drift
Feature drift
Input metadata drift
Prediction drift
Embedding drift
Output metadata drift

Appendix

Quality measures

Area under ROC

The Area under ROC is plotted parametrically as the [True positive rate](#) versus the [False positive rate](#) with respect to a threshold T .

Area under PR

Area under Precision Recall gives the total for both [Precision + Recall](#). Precision (P) is defined as the number of true positives (Tp) over the number of true positives plus the number of false positives (Fp)

Formula

$$\text{Precision} = \frac{\text{number of true positives}}{(\text{number of true positives} + \text{number of false positives})}$$

Recall (R) is defined as the number of true positives (Tp) over the number of true positives plus the number of false negatives (Fn).

$$\text{Recall} = \frac{\text{number of true positives}}{(\text{number of true positives} + \text{number of false negatives})}$$

Appendix

Quality measures

Accuracy

Base accuracy is calculated from the training data. It is the percentage of predictions that the model got correct when tested against the training data.

True positive rate (TPR)

The True positive rate is calculated by the following formula:

Formula

$$\text{TPR} = \frac{\text{number of true positives}}{(\text{number of true positives} + \text{number of false negatives})}$$

False positive rate (FPR)

The false positive rate is calculated as the total number of false positives divided by the number of false positives and the number of true negatives.

$$\text{FPR} = \frac{\text{number of false positives}}{(\text{number of false positives} + \text{number of true negatives})}$$

Appendix

Quality measures

Recall

Recall (R) is defined as the number of true positives (Tp) over the number of true positives plus the number of false negatives (Fn).

Formula

$$\text{Recall} = \frac{\text{number of true positives}}{(\text{number of true positives} + \text{number of false negatives})}$$

Precision

Precision (P) is defined as the number of true positives (Tp) over the number of true positives plus the number of false positives (Fp).

Formula

$$\text{Precision} = \frac{\text{number of true positives}}{(\text{number of true positives} + \text{number of false positives})}$$

Appendix

Quality measures

F1-Measure

The F1-Measure is the weighted harmonic average, or mean, of precision and recall.

Formula

$$F1 = 2 * \frac{(\text{precision} * \text{recall})}{(\text{precision} + \text{recall})}$$

Logarithmic loss

For a binary model, Logarithmic loss is calculated by using the following formula:

Formula

$$-(y \log(p) + (1-y) \log(1-p))$$

where p = true label and y = predicted probability

For a multi-class model, Logarithmic loss is calculated by using the following formula:

$$-\sum_{c=1}^M Y_{o,c} \log(P_{o,c})$$

where $M > 2$, p = true label, and y = predicted probability

Appendix

Fairness measures

Fairness

The fairness metric used in Watson OpenScale is disparate impact, which is a measure of how the rate at which an unprivileged group receives a certain outcome or result compares with the rate at which a privileged group receives that same outcome or result.

Formula

$$\text{Disparate impact} = \frac{(\text{num_positives(privileged=False)} / \text{num_instance(privileged=False)})}{(\text{num_positives(privileged=True)} / \text{num_instance(privileged=True)})}$$

Appendix

Drift measures

Drop in accuracy

Watson OpenScale analyzes each transaction to estimate if the model prediction is accurate. If the model prediction is inaccurate, the transaction is marked as drifted. The Estimated accuracy is then calculated as the fraction of non-drifted transactions to the total number of transactions analyzed. The Base accuracy is the accuracy of the model on the test data. Watson OpenScale calculates the extent of the drift in accuracy as the difference between Base accuracy and Estimated accuracy. Further, Watson OpenScale analyzes all the drifted transactions; and then, groups transactions based on the similarity of each feature's contribution to the drift in accuracy. In each cluster, Watson OpenScale also estimates the important features that played a major role in the drift in accuracy and classifies their feature impact as large, some, and small.

Drop in data consistency

Watson OpenScale analyzes each transaction for data inconsistency, by comparing the transaction content with the training data patterns. If a transaction violates one or more of the training data patterns, the transaction is marked as drifted. Watson OpenScale then estimates the magnitude of data inconsistency as the fraction of drifted transactions to the total number of transactions analyzed. Further, Watson OpenScale analyzes all the drifted transactions; and then, groups transactions that violate similar training data patterns into different clusters. In each cluster, Watson OpenScale also estimates the important features that played a major role in the data inconsistency and classifies their feature impact as large, some, and small.

Appendix

Drift measures

Estimated accuracy

Estimated accuracy is the accuracy score at runtime estimated by Watson OpenScale. As part of drift monitor configuration, Watson OpenScale trains a drift detection model that identifies when the original model is likely to provide an incorrect response to a transaction. As the original model receives a new transaction, the transaction is evaluated by the drift model. If the drift model believes that the model likely provided an incorrect response, the transaction is identified as a drifted transaction. The Estimated accuracy is then calculated as the fraction of non-drifted transactions to the total number of transactions analyzed.

Formula

$$\text{Estimated Accuracy} = \frac{\text{Number of non-drifted transactions}^*}{\text{Total number of transactions}}$$

*determined by the Watson OpenScale drift model

Base Accuracy

This is calculated from the training data. It is the percentage of predictions that the model got correct when tested against the training data.

Appendix

Drift v2 measures

Output drift

Measures the change in the model confidence distribution.

Model quality drift

Measures the drop in accuracy by comparing the estimated runtime accuracy to the training accuracy. Ground truth labels are not required.

Feature drift

Measures the change in value distribution for important features.

Input Metadata drift

Measures the change in the distribution of various input metadata e.g. input token count, input character count, etc.

Output Metadata drift

Measures the change in the distribution of various output metadata e.g. output token count, output character count, etc.

Embedding drift

Measures the percentage of transactions marked as outliers based on cosine and euclidean distance based on the embeddings of the input, output, individual variables.